

FIELD THEORY PART-II

Phase 10: PYQ Writing and Final Revision

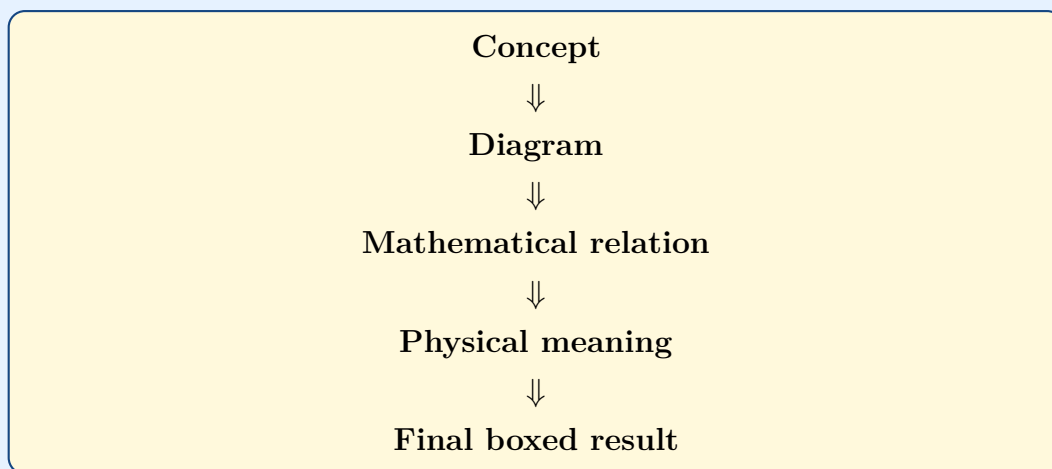
Converting knowledge into semester answer performance

Academic Article Style Notes for Semester Preparation

How to Use This Phase

This phase is made for semester preparation from a very basic level. Read the concept first, then understand the diagram, then learn the derivation pattern, and finally practise the exam-style writing block.

The recommended writing order is:



Contents

1	Aim of Phase 10	3
2	How to Select Questions	3
3	Standard Long Answer Structure	3
4	Standard Short Note Structure	4
5	Must-Write Diagrams	4
6	Formula Sheet	4
7	Final 7-Day Practice Schedule	5
8	Final Mistake Checklist	5
9	Final Exam Strategy	6

1 Aim of Phase 10

Main Goal

This phase converts knowledge into semester-paper performance. It contains final answer-writing strategy, PYQ priority, time management and revision checklist.

2 How to Select Questions

Priority Rule

In the exam, choose questions according to confidence and repetition.

Question Selection Priority

Choose in this order:

1. EM wave equation and plane wave.
2. Curl derivation with physical significance.
3. Stokes' theorem and Ampere's law.
4. Modified Ampere law and Faraday law.
5. Magnetic boundary condition.
6. Poynting theorem and polarization.
7. Biot-Savart numerical if geometry is clear.

3 Standard Long Answer Structure

Write Every Long Answer Like This

Definition



Diagram



Derivation



Physical meaning



Final boxed result

4 Standard Short Note Structure

Short Note Pattern

For short notes write:

1. Definition.
2. Formula.
3. Unit if required.
4. Physical meaning.
5. One example or application.

5 Must-Write Diagrams

Diagram List

Prepare these diagrams by hand:

1. Small rectangular loop for curl.
2. Open surface for Stokes' theorem.
3. Wire with circular magnetic field for Ampere law.
4. Boundary between two magnetic media.
5. Capacitor for displacement current.
6. Circular electric field for Faraday law.
7. Plane EM wave propagation.
8. Poynting vector direction.
9. Polarization diagrams.

6 Formula Sheet

Compulsory Formulas

$$\nabla \times \mathbf{H} = \mathbf{J}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \cdot \mathbf{J} = 0$$

$$\mathbf{B} = \mu \mathbf{H}$$

$$\mathbf{D} = \epsilon \mathbf{E}$$

$$\mathbf{J} = \sigma \mathbf{E}$$

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$\nabla^2 \mathbf{H} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

7 Final 7-Day Practice Schedule

Day	Work	Output
1	Curl and divergence	One full derivation each
2	Stokes and Ampere law	Two long answers ready
3	Biot–Savart and boundary condition	One numerical and one theory answer
4	Modified Ampere and Faraday	Two long answers with diagrams
5	EM wave equation and plane wave	One complete 10-mark answer
6	Poynting and polarization	One theorem and short notes
7	Full PYQ simulation	Attempt any three long questions and four short notes

8 Final Mistake Checklist

Avoid These Mistakes

- Do not write $\nabla \cdot \mathbf{B}$ as non-zero.
- Do not forget the minus sign in Faraday law.
- Do not forget displacement current in time-varying Ampere law.
- Do not mix $\mathbf{B}$ and $\mathbf{H}$.
- Do not draw tiny unreadable diagrams.
- Do not write only formulas without explanation.
- Do not skip units in numerical answers.

9 Final Exam Strategy

During Exam

1. Read all Part-II questions first.
2. Mark the familiar repeated topics.
3. Start with the cleanest long answer.
4. Put diagrams before derivations.
5. Box final formulas.
6. Leave space between steps.
7. Use short notes to secure marks.

End Condition

Preparation is complete when you can write these five answer families:

Curl–Divergence–Stokes

Ampere–Biot Savart–Boundary Condition

Modified Ampere–Faraday

EM Wave Equation–Plane Wave

Poynting–Polarization–Short Notes

10 Final Self-Test for This Phase

Instructions

Answer these without looking at the notes. If you cannot answer a point, revise that section again before moving to the next phase.

1. Can I state the main definition of this phase?
2. Can I draw the main diagram neatly without overlap?
3. Can I explain the physical meaning in two or three lines?
4. Can I write the final formula with correct symbols?
5. Can I connect this topic with at least one past-year question pattern?
6. Can I write a semester-style answer with definition, diagram, derivation and final result?

Completion Rule

This phase is complete only when the concept, diagram, formula and answer-writing pattern are all ready.